

Yatharth Singh Chouhan

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Professional Summary

Software Development Engineer with experience in developing scalable backend systems and improving system performance and reliability, **REST APIs**, and data pipelines using **Python**, **Django**, and **Flask**. Proficient in **semantic search**, **cosine similarity**, **NLP embeddings**, and **AI integration** with experience processing 100K+ real-world records. Strong **problem-solving** foundation in DSA, OOPs, and Operating System ; writes **unit-tested**, production-grade code across the full **software development lifecycle** in Agile environments.

Professional Experience

Techritzy Private Limited, Indore, India

Nov 2025 – Present

Software Developer Intern

- Architected and developed scalable backend services using **Flask** and **Django**, building robust **RESTful APIs** with JSON-based communication to support efficient and reliable microservices architecture.
- Designed and implemented **30+** RESTful API endpoints using **Django REST Framework** (DRF) to support core application features and enhance backend performance.
- Implemented **SSO** with **SAML 2.0** and **OAuth 2.0 (Google Sign-In)**- based **authentication** and **authorization** mechanisms to enable secure and seamless user access.
- Built secure authentication and authorization using **JWT** and **RBAC**, ensuring controlled access and adherence to clean code practices and standard design patterns.
- Worked on **SQL** queries and database schemas to improve system reliability, while collaborating with the team throughout the **SDLC** and using **Git** for version control.
- Worked closely with senior engineers to **debug**, test, and optimize third-party integrations, demonstrating strong **problem-solving** skills and contributing to technical documentation.

AARUUSH, SRM Institute of Science and Technology, Chennai

Apr 2024 – May 2025

Organiser, Team Outreach

- Led coordination between **5+ college teams** and **10+ stakeholders** to organize technical events with **500+ attendees**, demonstrating ownership and leadership in complex, large-scale environments.
- Head of Technology for the global **NASA Space Apps Challenge** (multi-college), managing a tech team and overseeing end-to-end technical operations.

Projects

Review Based Restaurant Recommendation System | *Flask, AI Embeddings, MySQL, Cosine Similarity* [GitHub](#)

- Addressed limitations of rating-based recommendation systems by leveraging **user review data**; built web scraping pipelines to collect and process **100K+ Zomato reviews**, enabling structured input for AI-driven analysis.
- Converted review corpus into **text embeddings** and implemented **semantic search using cosine similarity** to surface contextually relevant recommendations beyond keyword matching.
- Wrote **unit tests covering best, worst, and average cases** to validate recommendation accuracy, edge case handling, and system reliability.

Bulk Transaction Engine | *Django, DRF, Redis, Celery, MySQL*

[GitHub](#)

- Built modular backend using **Django REST Framework** for product, cart, and order services with **JWT auth** and **RBAC**; maintained ACID-compliant transactional integrity across workflows.
- Integrated **Celery** for **async task processing** and **Redis caching** to enhance API response time and system scalability.

CineBook | *Flask, MySQL, ACID Transactions, Python*

[GitHub](#)

- Built a **concurrent seat booking engine** preventing race conditions via ACID transactions; modeled relational schema for fault tolerance under concurrent load.
- Implemented **seat availability filtering by show and time slot**, **role-based access control** (admin vs user), and **payment/booking summary flow** with transactional integrity across the reservation lifecycle.

Technical Skills

Languages	Python 3 , C++ , SQL
Backend	Django, Flask, Django REST framework (DRF), JWT Authentication, Authorization, SSO
AI / Search	Semantic Search, Cosine Similarity, Text Embeddings, NLP, Web Scraping
Databases & Cache	MySQL, Redis, ACID Transactions, Query Optimization
Cloud & DevOps	AWS EC2, Git, Github, Linux
CS Fundamentals	Data Structures & Algorithms, OOPs, OS Concepts, Design Patterns
Testing	Unit Testing, pytest

Certifications

AWS Academy Machine Learning Foundations | Cisco Networking Academy | Udemy Web Development

Education

Bachelor of Technology in Computer Science and Engineering

2021 – 2025

SRM Institute of Science and Technology

CGPA: 7.49 / 10.0